

Kellanova / Mars

Fairness Opinion Replication

NYSE: K · acquired by Mars · US\$ millions · per-share values in US\$ · as of DEFM14A filed 26 Sep 2024 · valuation date 30 Jun 2024

A replication of the fairness-opinion analyses Goldman Sachs and Lazard delivered to Kellanova's board on Mars's US\$83.50 cash offer, rebuilt from the inputs disclosed in the definitive proxy of 26 September 2024: management projections, discount rates, multiple and premia ranges, and the undisturbed price. Each rebuilt range is tested against the range the proxy reports.

What this is

- Fully diluted shares and adjusted net debt back-solved from a single disclosed range (Goldman's precedent transactions), then held fixed for every other analysis so that each is a genuine test.

How it is built

- Goldman: DCF with a terminal multiple, present value of future share price with dividends, premia paid on two bases, precedents, and reference trading multiples. Lazard: DCF with perpetuity growth, EBITDA and P/E comparables, precedents, premia, analyst targets and trading range.
- A football field placing the offer against all twelve ranges, with the deviation of each rebuilt bound from the proxy.

Contents

[Summary](#) — [the deal, the replication, the reading](#)
[Inputs](#) — [everything disclosed; the two back-solved figures](#)
[Goldman Sachs](#) — [five analyses rebuilt](#)
[Lazard](#) — [four analyses rebuilt](#)
[Football Field](#) — [twelve ranges against US\\$83.50](#)
[Checks](#) — [fifteen tests; must read MODEL OK](#)

Key outputs (live)

Offer per share (US\$)	83.50
Premium to undisturbed	32.6%
EV / 2024E EBITDA at the offer	16.0x
GS DCF rebuilt: low	67.79
GS DCF rebuilt: high	83.19
Largest deviation from the proxy, core analyses (US\$)	2.52

Colour code

Blue = hard-coded input
Black = formula
Green = link to another sheet
Accent fill = scenario selector / key levers

Sources

Kellanova definitive proxy statement (DEFM14A), SEC accession 0001193125-24-226970, filed 26 September 2024: 'Certain Financial Projections', 'Opinion of Goldman Sachs & Co. LLC', 'Opinion of Lazard Frères & Co. LLC', and the record-date share count.

Dividends per share after 2024 and the split of 2024 cash flow either side of 30 June are stated inputs; every other figure is quoted from the proxy or derived from it as labelled.

The exercise is a replication of disclosed analyses for learning. It expresses no view on either advisor's judgement, and it is not advice.

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Everything the proxy discloses, and the two figures back-solved from it

TRANSACTION

Merger consideration per share (cash)	83.50	
Undisturbed closing price, 2 August 2024	62.98	Also the 52-week high on that date
Premium to undisturbed	32.6%	
Shares outstanding at the record date (millions)	344.68	

MANAGEMENT PROJECTIONS (proxy, 'Certain Financial Projections')

	2024E	2025E	2026E	2027E
Net sales	12,782	13,366	13,670	14,098
Adjusted EBITDA	2,235	2,401	2,553	2,741
Net operating profit after tax	1,322	1,506	1,595	1,705
Unlevered free cash flow	1,102	1,348	1,359	1,533
Adjusted earnings per share	3.70	3.99	4.26	4.63
Dividends per share (stated: 2024 run-rate, then +3.5% a year)	2.28	2.36	2.44	

BACK-SOLVED FROM GOLDMAN'S PRECEDENT-TRANSACTION RANGE

Goldman applied 12.1x–21.5x to FY1 (2024E) EBITDA and reported \$58.81–\$118.63 per share. Two equations, two unknowns.

Fully diluted shares (millions) = $(21.5 - 12.1) \times \text{FY1 EBITDA} / (118.63 - 58.81)$	351.20
Adjusted net debt = $12.1 \times \text{FY1 EBITDA} - 58.81 \times \text{diluted shares}$	6,389
check: 21.5x reproduces the top of the range	118.63

About 2% above the record-date basic count — the treasury-stock dilution one would expect

Financial net debt plus factored receivables and minorities less unconsolidated investments, as the proxy defines it

HALF-YEAR CONVENTION

Share of 2024E unlevered free cash flow falling after 30 June 2024	50.0%	The proxy discounts cash flow from 1 July 2024; the split is not disclosed
Mid-year discount period for H2 2024 (years from 30 June 2024)	0.25	

Every number above the back-solved section is quoted from the proxy. Diluted shares and adjusted net debt are the only derived inputs, and they are derived from a single disclosed analysis so that every other analysis is a genuine test.

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Five analyses rebuilt from the disclosed inputs, each against the range the proxy reports

1. ILLUSTRATIVE DISCOUNTED CASH FLOW — WACC 6.0–7.0%, terminal 12.0–14.0x 2027E EBITDA, mid-year, from 30 June 2024

	H2 2024	2025E	2026E	2027E
Unlevered free cash flow	551	1,348	1,359	1,533
Discount period (years, mid-period)	0.25	1.00	2.00	3.00
Low: 7.0% WACC, 12.0x terminal (US\$ per share)	67.79			
High: 6.0% WACC, 14.0x terminal	83.19			
Proxy: \$68.42 – \$83.82	68.42 – 83.82			
Implied perpetuity growth at 12.0x and 6.0%: $g = (TV \times r - CF_{2027}) / (TV + CF_{2027})$	1.3%			
Implied perpetuity growth at 14.0x and 7.0%	2.9%			

Proxy: 'implied perpetuity growth rates ranging from 1.0% to 2.6%'

2. PRESENT VALUE OF FUTURE SHARE PRICE — 11.5–14.5x NTM EBITDA at each year-end, dividends added, discounted at 6.9% cost of equity

Value at year-end, then discounted to 30 June 2024	YE 2024	YE 2025	YE 2026
At 11.5x	60.65	63.37	66.56
At 14.5x	80.49	83.10	86.37
Rebuilt range — low	60.65		
Rebuilt range — high	86.37		
Proxy: \$60.93 – \$90.01	60.93 – 90.01		

3. PREMIA PAID — US targets over US\$20bn since 2008, strategic buyers

14% over undisturbed	71.80		
32% over undisturbed	83.13		
Proxy: \$71.80 – \$83.13	71.80 – 83.13		
7% over the 52-week high	67.39		
23% over the 52-week high	77.47		
Proxy: \$67.39 – \$77.47	67.39 – 77.47		
Where the offer sits: premium to undisturbed versus the 75th percentile	0.6%		

The offer is above every precedent premium in the set: Mars paid more than the top quartile

4. PRECEDENT TRANSACTIONS — 12.1–21.5x FY1 EBITDA (the calibrating analysis)

12.1x 2024E EBITDA	58.81		
21.5x 2024E EBITDA	118.63		
Proxy: \$58.81 – \$118.63 (reproduced by construction)	58.81 – 118.63		
Multiple of 2024E EBITDA the offer implies	16.0x		
Multiple of 2025E EBITDA the offer implies	14.9x		
Median of the eleven precedents (proxy)	16.2x		

Wrigley/Mars 2008 at 17.6x is in the set

5. TRADING COMPARABLES — 'for reference purposes only'

Median EV/NTM EBITDA, snacking (Hershey, Mondelēz, PepsiCo)	14.8x
Median EV/NTM EBITDA, centre of store (Campbell, Conagra, General Mills,	9.9x
Offer multiple of NTM (2025E) EBITDA against the snacking median	0.07

Turns of EBITDA above the best-rated peer group

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Lazard's analyses rebuilt with the same back-solved balance sheet

1. DISCOUNTED CASH FLOW — WACC 6.25–6.75%, perpetuity growth 1.5–2.0%, mid-year, from 30 June 2024

Low: 6.75% WACC, 1.5% growth	61.07
High: 6.25% WACC, 2.0% growth	78.76
Proxy: \$69.60 – \$79.10	69.60 – 79.10

Close to 2027E NOPAT (US\$1,705m): Lazard appears to have normalised the terminal year, removing the working-capital build in 2027E free cash flow

Terminal free cash flow that reproduces Lazard's \$69.60 low end at 6.7	1,728
as a multiple of 2027E unlevered free cash flow	1.1x
Terminal value at the midpoint as a multiple of 2027E EBITDA	12.0x

Lazard's growth method and Goldman's multiple method should meet in the middle; this shows whether they do

2. SELECTED PUBLIC COMPANIES — 11.0–13.5x 2025E EBITDA and 14.0–19.0x 2025E EPS

11.0x 2025E EBITDA	57.01
13.5x 2025E EBITDA	74.10
Proxy: \$57.20 – \$73.80	57.20 – 73.80
14.0x 2025E adjusted EPS	55.86
19.0x 2025E adjusted EPS	75.81
Proxy: \$55.90 – \$75.70	55.90 – 75.70
Observed: snacking 2025E EBITDA low / high / median	14.3x / 14.9x / 14.5x
Observed: diversified grocery 2025E EBITDA low / high / median	9.1x / 12.1x / 9.4x

3. PRECEDENT TRANSACTIONS — 15.5–17.5x LTM adjusted EBITDA (eleven food deals since 2014; median 16.7x)

LTM adjusted EBITDA to 29 June 2024, back-solved from the disclosed r	2,125
15.5x LTM	75.58
17.5x LTM	87.68
Proxy: \$78.10 – \$90.20	78.10 – 90.20
Multiple of LTM EBITDA the offer implies	16.8x

Sits between 2023 actual and 2024E, as an LTM figure to mid-year should

4. OTHER ANALYSES (informational, not the basis of the opinion)

Premia paid: 20% over the 2 August close	75.58
Premia paid: 35% over the 2 August close	85.02
Proxy: \$75.60 – \$85.10	75.60 – 85.10
Analyst price targets, discounted one year at 7.0%: low	56.07
high	68.22
Post-separation trading range (since 2 October 2023): low	48.62
high	62.98

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Every range on one sheet against the US\$83.50 offer

RANGES — rebuilt, with the proxy's own figures alongside

Analysis	Rebuilt low	Rebuilt high	Proxy low	Proxy high	Offer vs proxy range	Max deviation
GS — discounted cash flow	67.79	83.19	68.42	83.82	Inside	0.63
GS — present value of future share price	60.65	86.37	60.93	90.01	Inside	3.64
GS — premia paid, undisturbed	71.80	83.13	71.80	83.13	Above	0.00
GS — premia paid, 52-week high	67.39	77.47	67.39	77.47	Above	0.00
GS — precedent transactions	58.81	118.63	58.81	118.63	Inside	-
Lazard — discounted cash flow	61.07	78.76	69.60	79.10	Above	8.53
Lazard — public companies, EBITDA	57.01	74.10	57.20	73.80	Above	0.30
Lazard — public companies, P/E	55.86	75.81	55.90	75.70	Above	0.11
Lazard — precedent transactions	75.58	87.68	78.10	90.20	Inside	2.52
Lazard — premia paid	75.58	85.02	75.60	85.10	Inside	0.08
Lazard — analyst targets	56.07	68.22	56.07	68.22	Above	-
Lazard — post-separation trading	48.62	62.98	48.62	62.98	Above	-
Ranges the offer exceeds entirely	7					
Ranges the offer falls inside	5					
Ranges the offer falls below	0					
Largest deviation between a rebuilt bound and the proxy's (US\$ per	8.53					
excluding the future-share-price analysis and the Lazard DCF low end (	2.52					
Lazard DCF: deviation at the high end alone	0.34					
Highest rebuilt bound across the opinion-basis analyses (GS 1–4, Lazard	118.63					

The offer sits at the top of Goldman's discounted cash flow range and above Lazard's, at or above the top of the premia ranges, above every trading-multiple range and inside only the precedent-transaction ranges. That is what a fairness opinion on a full-price strategic deal looks like: the intrinsic analyses bracket the price, the market-based ones sit beneath it, and the precedents — which include the buyer's own Wrigley deal at 17.6x — are the one place a higher number can be found.

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What the proxy discloses, what was rebuilt, and what it says about the price

THE DEAL

Offer per share (cash)	83.50
Premium to the undisturbed price	32.6%
Equity value at the offer (US\$ m)	29,326
Enterprise value at the offer (US\$ m)	35,715
EV / 2024E EBITDA · EV / 2025E EBITDA · EV / LTM EBITDA	16.0x · 14.9x · 16.8x

THE REPLICATION

Goldman DCF: rebuilt vs proxy	67.79–83.19 vs 68.42–83.82
Lazard DCF: rebuilt vs proxy	61.07–78.76 vs 69.60–79.10
Lazard EBITDA comps: rebuilt vs proxy	57.01–74.10 vs 57.20–73.80
Largest deviation of any rebuilt bound from the proxy, core analyses (US\$)	<u>2.52</u>
Ranges the offer exceeds / sits inside / sits below	<u>7 / 5 / 0</u>

• The proxy discloses enough to rebuild almost every analysis to within a dollar or two a share: projections, discount rates, multiple ranges, premia ranges and the undisturbed price. The two undisclosed inputs — diluted shares and adjusted net debt — fall out of a single disclosed range and then reproduce the rest. Goldman's DCF lands within 63 cents, Lazard's comparables within 30 cents, the premia to the cent. The one miss is the low end of Lazard's DCF, which needs a terminal cash flow close to NOPAT rather than 2027E free cash flow — the sheet says so and back-solves it.

• The offer is a full price by every intrinsic measure. It sits at the top of Goldman's DCF range, above Lazard's, at or above the top of the premia ranges, and roughly 3–4 turns of EBITDA above the best-rated listed peers. Only the precedent-transaction ranges, anchored by 2015–2018 food deals and Mars's own Wrigley acquisition, leave room above it.

• The two advisors agree because they were given the same projections and the same balance sheet; their methods differ (a terminal multiple against a perpetuity growth rate) and land within a few dollars of each other. A reader should treat that as consistency of inputs, not independent confirmation.

• What cannot be rebuilt is the judgement: why 12–14x rather than 11–15x, why 14–32% rather than the full quartile spread. The proxy gives the ranges and the phrase 'professional judgment and experience'. A fairness opinion is a set of defensible ranges that bracket a price the board has already negotiated; this replication shows how tightly they bracket it.

INTEGRITY CHECKS

Back-solved diluted shares are within 0–5% above the record-date basic count	TRUE
Back-solved net debt reproduces both ends of Goldman's precedent range	TRUE
Goldman DCF rebuilt within US\$2.5 of the proxy at both ends	TRUE
Goldman's implied perpetuity growth range brackets the proxy's 1.0–2.6%	TRUE
Goldman premia-paid ranges reproduce to the cent	TRUE
Goldman present-value-of-future-share-price rebuilt within US\$5 at both ends (dividend path	TRUE
Lazard DCF high end rebuilt within US\$2.5; low end not reproducible on 2027E free cash flow	TRUE
Terminal cash flow needed for Lazard's low end lies between 2027E free cash flow and 20	TRUE
Lazard EBITDA comps rebuilt within US\$2.5 at both ends	TRUE
Lazard P/E comps reproduce within 20 cents	TRUE
Lazard premia reproduce within 10 cents	TRUE
Back-solved LTM EBITDA lies between 2023 actual (about US\$2,150m) and 2024E	TRUE
Offer lies inside Goldman's DCF range and above Lazard's, as disclosed and as rebuilt	TRUE
Offer exceeds every trading-multiple range	TRUE
Football-field counts sum to twelve analyses	TRUE

MODEL OK

The checks are the point of the project: each one asks whether an analysis rebuilt from the proxy's disclosed inputs lands on the range the advisor reported. A tolerance of US\$2.50 a share allows for the undisclosed H2-2024 cash-flow split and rounding; the premia and P/E analyses have no such freedom and must reproduce almost exactly.